

Video Outline: Master CSS Color Models in 5 Minutes

Target Duration: ~4:30 – 5:00 minutes

Topic: CSS Color Models (Hex, RGB, RGBA, HSL, HSLA) & CSS Variables

1. Introduction (0:00 – 0:40)

Hook (0:00 – 0:15): Brief visual hook showing a sleek UI switching themes dynamically, highlighting why understanding CSS color systems makes styling effortless.

Overview (0:15 – 0:30): Quick breakdown of what will be covered: Hex, RGB/RGBA, HSL/HSLA, and how CSS variables tie them together.

Goal (0:30 – 0:40): Set the objective—knowing exactly which color format to choose for any design requirement.

2. Concept Explanation (0:40 – 2:10)

Hex Codes (0:40 – 1:00):

Explanation of the #RRGGBB base-16 format.

When to use: Standard static colors, quick copy-pasting from design files.

RGB & RGBA (1:00 – 1:35):

Explanation of rgb(red, green, blue) values from 0–255.

Transition to rgba() for controlling alpha/opacity (0.0 to 1.0).

When to use: Overlays, drop shadows, and semi-transparent elements.

HSL & HSLA (1:35 – 2:10):

Explanation of Hue (0–360°), Saturation (0–100%), Lightness (0–100%), and Alpha.

When to use: Creating color palettes, hover states, and dark mode toggles intuitively.

3. Live Code Walkthrough (2:10 – 4:10)

Setup (2:10 – 2:30): Show a simple card component styled with standard Hex codes.

Refactoring with HSL & Opacity (2:30 – 3:15):

Convert a Hex background to hsl().

Demonstrate adding an overlay/shadow using rgba() or hsla() (or modern slash syntax hsl(h s l / a)).

Show how tweaking Lightness in HSL instantly creates a hover state without searching for a new Hex code.

Integrating CSS Variables (3:15 – 4:10):

Define primary color variables in :root (--primary-color, --bg-alpha).

Show how combining CSS variables with HSL makes theming and dark mode swapping instant across an entire stylesheet.

4. Summary and Conclusion (4:10 – 5:00)

Quick Recap (4:10 – 4:35):

Hex: Quick & universal.

RGB/RGBA: Best for digital color channels & basic transparency.

HSL/HSLA: Human-readable, ideal for palettes & hover states.

CSS Variables: Essential for maintainable design systems.

Call to Action & Outro (4:35 – 5:00): Encourage viewers to experiment with HSL + CSS variables in their next project, with standard closing prompts.